

VEDAANG CHOPRA

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MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

Research: Production agentic systems with execution-grounded verification – Multi-agent orchestration & tool use – Cost-aware VLM routing

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (*Machine Learning Specialization*), **GPA: 4.0/4.0**

Aug 2024 – Dec 2026

– **Coursework:** Deep Learning, Deep RL, ML Security, Large & Vision Language Models, Agentic AI, Systems for AI

Maharaja Surajmal Institute of Technology

New Delhi, India

B.Tech. in Information Technology, CGPA: 8.8/10.0

Aug 2016 – Aug 2020

RESEARCH EXPERIENCE

Georgia Institute of Technology – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- Building a closed-loop agentic pipeline for parametric CAD code generation: an LLM generates CadQuery programs, an execution environment returns compile and geometry feedback, and a LangGraph-orchestrated repair loop iteratively corrects errors – framing synthesis as inference-time search with verifiable geometric reward rather than single-pass generation.
- Designing geometric verification components that score compile validity, parametric 3D correctness (Chamfer/Hausdorff distances on STL and B-Rep representations), and repair trajectory quality; maintaining a 202-test evaluation harness with VLM-based visual judgment and RAG-augmented prompt construction across a 16-module system.

Georgia Institute of Technology – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **ATHENA** (Jan–May 2025): Built a multi-agent screenplay generation system with supervisor-worker orchestration using LangGraph Command routing; coordinated 5 specialized agents (ideation, character design, world building, story generation, scene breakdown) with reflection-based quality critique and dynamic plan modification; evaluated with BLEU-4 and CLIPScore metrics.
- **SHAstra** (Jan 2026–Present): Designing an agentic workflow retrieval and adaptation framework that enables compositional reuse of prior tool-call sequences for new long-horizon tasks under cost/latency constraints; implementing a LangGraph-based planner with Pydantic state and a component registry with pluggable storage.
- **ARTEMIS** (Aug 2025–Present): Developing a cost-aware routing system for Vision-Language Models that dynamically selects the optimal VLM per request; trained a neural multi-task router with SLA-aware load balancing, serving six VLM backends via vLLM across VQA, OCR, captioning, and reasoning tasks.

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- Led development of an agentic RAG diagnostics system: LLM plans multi-step tool calls over network telemetry, invokes APIs, and verifies hypotheses via structured function calling, producing autonomous root-cause analysis – reduced mean resolution time 70% (hackathon prototype to production deployment).
- Scaled OpenSearch ingestion pipelines from 50 to 2,000 events/sec using Golang while maintaining reliability and observability; integrated ML models into backend services with Python and Go across distributed systems and telemetry infrastructure.
- Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection using density-based analysis; reduced manual troubleshooting 75%.

PUBLICATIONS & PATENTS

Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection

Ontology-based Text Classification — Published: CEUR Workshop Proceedings Vol. 2786

2020

TECHNICAL SKILLS

Agentic AI: LangGraph, LangChain, Agentic RAG, Tool/Function Calling, Multi-Agent Orchestration, Supervisor-Worker Patterns, Planning & Reasoning, LLM-as-a-Judge, Structured Generation (Pydantic)

AI / ML: PyTorch, Hugging Face Transformers, Deep Learning, CNNs, Vision Transformers, NLP, Computer Vision, Scikit-Learn, NumPy, Pandas

LLMs / VLMs: vLLM, VLM Routing & Evaluation, CLIP, SBERT, FAISS, Inference Optimization, Model Evaluation & Benchmarking, Cross-Modal Retrieval

Reinforcement Learning: PPO, DQN, Reward Shaping, Curriculum Learning, Self-Play (academic projects)

Systems: Python, Go, C/C++, SQL/PostgreSQL, Redis, OpenSearch/Elasticsearch, Docker, Linux, Git, ONNX Runtime, FastAPI, Azure, W&B, TensorBoard